MIL-S-45869(PA)
9 June 1975
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

SWITCH, PRESSURE, FOR T4016(T310) TEST SET

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of Switch, Pressure for use on the T-310 Baroswitch Test Set which is used with the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1167	- Ammunition Data Card.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

ORDNANCE CORPS

8805838 - Switch, Pressure.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

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3.2 Assembly. The Pressure Switch shall comply with all requirements specified on the drawing (dwg) 8805838, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Environmental testing. The switch shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The switch shall also comply with 3.4.1 where applicable before and after each environmental test. Testing shall be as specified in 4.4.1. Tests shall be conducted in the order specified.

3.4.1 Leakage. The product of the leakage rate expressed in millimeters of mercury (mm Hg) per minute and the effective volume expressed in cubic inches shall not exceed a value of 20 when tested as specified in 4.4.1.1.

3.4.2 Operating pressure. The pressure switch shall be capable of operating at any pressure between the limits of 137.4 ± 4 mm Hg to 848 ± 4 mm Hg absolute when tested as specified in 4.4.1.2.

3.4.3 Transportation vibration. The pressure switch shall show no evidence of damage and shall operate as specified in 4.4.1.2 after subjection to a vibration of 10 to 55 to 10 Hertz (Hz) at a double amplitude of 0.03 inch when tested as specified in 4.4.1.3.

3.4.4 Life. The pressure switch shall be capable of of continuous operation as specified in 4.4.1.2 for 15,000 cycles when tested as specified in 4.4.1.4.

3.4.5 High temperature. The pressure switch shall be capable of operating after being stabilized (see 6.4) at a temperature of 130 ± 5 degrees Fahrenheit (°F) when tested as specified in 4.4.1.5.

3.4.6 Low temperature. The pressure switch shall be capable of operating after being stabilized at a temperature of 0 ± 5°F when tested as specified in 4.4.1.6.

3.4.7 Humidity. The pressure switch shall show no evidence of deterioration or damage such as obliterated markings or distorted, cracked, chipped, peeled, swollen, or eroded components and shall meet the requirements of 3.4.2 after being tested as specified in 4.4.1.7.

3.5 Workmanship.

3.5.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.5.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

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3.5.3 Burrs. Parts shall be free from burrs.

3.5.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.5.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8805838) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Leakage (see 3.4.1)		X
3. Operating pressure (see 3.4.2)		X
4. Transportation vibration (see 3.4.3)		X
5. Life (see 3.4.4)		X
6. High temperature (see 3.4.5)		X
7. Low temperature (see 3.4.6)		X
8. Humidity (see 3.4.7)	X	

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Switch, pressure (see dwg 8805838).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Pitch diameter of pressure port threads	Gage
	102. Minor diameter of pressure port threads	Gage
	103. Hardware loose or missing	Visual/Manual
	104. Bent or broken pin or connector	Visual
	105. Foreign matter in pressure port or connector	Visual
	106. Cracks or dents in cover	Visual
	107. Protective finish damaged exposing base metal	Visual
	108. Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pins)	Visual
	109. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Marking illegible, missing, or incorrect	Visual

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202. Evidence of poor workmanship (see 3.5).....Visual

*In the event pin(s) do not appear to be within 1/16 inch in height relative to the adjacent pins, Gage (see 4.3.4) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.063 inch max. from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface of all measurements of pin heights.

4.3.2.1.2 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation, packing, and packaging not in accordance with MIL-STD-1169	Visual
	102. Marking not in accordance with MIL-STD-1169	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.5)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Leakage	Major
b. Operating pressure	Major
c. Transportation vibration	Major
d. High temperature	Major
e. Low temperature	Major
f. Humidity	Major

Of these tests, f is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quanitity of items required by the contract.

4.3.3.1 Leakage and operating pressure. Each switch shall be tested in accordance with 4.4.1.1 and 4.4.1.2. If any item fails to comply with 3.4.1 and 3.4.2, the item shall be rejected and removed from the lot.

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4.3.3.2 Transportation vibration, high temperature, low temperature and humidity. Three switches shall be selected at random from each lot. The sample shall be tested in accordance with 4.4.1.3, 4.4.1.5, 4.4.1.6 and 4.4.1.7. For lots of 301 or more, MIL-STD-105 with an Acceptable Quality Level (AQL) of 0.25 percent will be used. If any switch fails to comply with 3.4.3, 3.4.5, 3.4.6 and 3.4.7, the lot shall be rejected.

4.3.4 Acceptance inspection equipment. Inspection and testing shall be performed with the acceptance test equipment, operating instructions and calibration instructions prepared by the contractor. The contractor shall obtain approval of such designs, operating instructions and special calibration instructions as specified in 6.3 prior to use on the contract.

4.3.4.1 Minor characteristics and tests. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Environmental testing (see 3.4).

4.4.1.1 Leakage. Evacuate the internal cavity pressure to 500 mm Hg less than the external pressure and observe the leakage rate with the use of a manometer or equivalent pressure measuring device. The product of the leakage rate expressed in mm Hg per minute and the effective volume expressed in cubic inches shall not exceed a value of 20. The observed time period shall be more than five minutes but less than ten minutes. The effective volume for the purposes of this test shall not exceed 30 cubic inches.

4.4.1.2 Operating pressure.

4.4.1.2.1 High altitude switch test. While monitoring the high altitude switch contacts (A, B, and C,), decrease the internal cavity pressure from ambient pressure to a pressure of 187.4 $\pm$ 4 mm Hg absolute. Contacts A and C shall remain closed and contacts B and C shall remain open. The pressure shall then be decreased until the internal cavity pressure reaches 137.4 $\pm$ 4 mm Hg absolute. Contacts B and C shall have closed and A and C shall have opened. The internal cavity pressure shall then be increased to 220 $\pm$ 4 mm Hg absolute. Contacts B and C shall have opened and A and C shall have closed.

4.4.1.2.2 Low altitude switch test. While monitoring the low altitude switch contacts (D, E, and F,), increase the internal cavity pressure to 816 + 4mm Hg absolute. Contacts D and F shall remain closed and D and E shall remain open.

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The internal cavity pressure shall be increased to 848 ± 4mm Hg absolute. Contacts D and F shall have opened and D and E shall have closed. The internal cavity pressure shall be decreased to 790 ± 4 mm Hg absolute. Contacts D and F shall have closed and D and E shall have opened.

4.4.1.3 Transportation vibration. The switch shall be secured to a vibration table and vibrated for 45 minutes in each of three mutually perpendicular planes through a frequency of 10 to 55 to 10 Hz at a double amplitude 0.03 inch. The frequency shall be varied uniformly from 10 to 55 to 10 Hz at approximately one cycle per minute. After completion of this test switch shall be tested as specified in 4.4.1.2.

4.4.1.4 Life test. The switch shall be operated for 15,000 cycles at any cycle rate between one and five cycles per minute. A complete cycle shall be defined as an operation in which the assembly is subjected to a pressure varying from 838 ± 14 mm Hg absolute (2000 to 3000 feet below sea level) to 162.4 ± 25 mm Hg absolute (34,000 to 40,500 feet above sea level) and back to 838 ± 14 mm Hg absolute during which time each electrical switch contact shall be capable of performing one make and one break of a 30 volt dc potential, 2.5 ampere, inductive load during each cycle.

4.4.1.5 High temperature. Place the switch in a test chamber and make suitable pressure and electrical connection. The relative humidity should be 20 percent or less. Raise the chamber temperature to 130 ± 5°F. and stabilize (see 6.4) the pressure switch at this temperature. While at this temperature the switch shall be tested as specified in 4.4.1.2.

4.4.1.6 Low temperature. Place the switch in a test chamber and make suitable pressure and electrical connection. The relative humidity at the beginning of the test shall be high enough to produce frozen condensate at 0°F. Stabilize (see 6.4) the temperature at 0 ± 5°F. and test the switch as specfiied in 4.4.1.2.

4.4.1.7 Humidity. The switch with all connections made and in normal position shall be placed in a test chamber. The humidity shall be maintained at 90 to 98 percent relative humidity throughout the following cycle. The temperature shall be raised to 130 ± 5°F within four hours maximum and maintained at this point for eight hours minimum. The temperature shall then be reduced to 86 ± 5°F within four hours maximum and maintained at this point for 21 hours minimum. Between the 8th and 12th hours after stabilization (see 6.4) at 86 ± 5°F, the switch shall be reduced to 68 ± 5°F within one hour maximum and maintained at this point for four hours minimum. The temperature shall than be raised to 86 ± 5°F within one hour maximum and maintained at this point for five hours minimum. At the conclusion of this test the switch shall be tested as specified in 4.4.1.2

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The switch shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Level C. The switch shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.2 Packing.

5.2.1 Levels A and C. The switch shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Data cards. Data cards shall be prepared in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4 and 5.4). Data specified in 4.3.4 and 5.4 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1178 dated 23 April 1959.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A086

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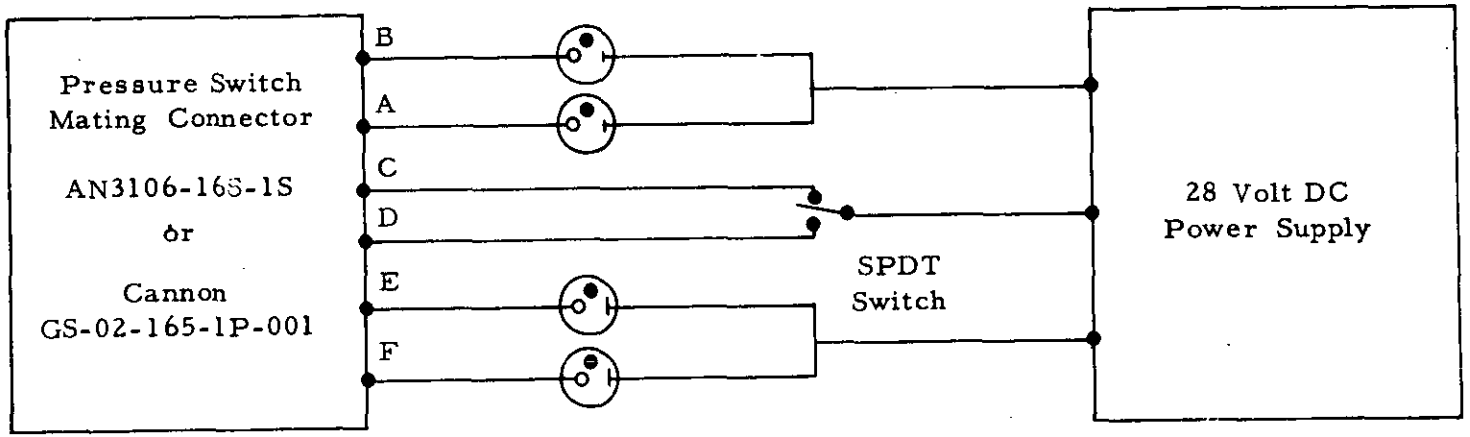


Figure 1. Pressure Switch Closure Test Circuit

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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